

SUMMARY

Robotics Engineering graduate, experienced across embedded systems, PLC programming, industrial robot programming, and custom PCB design - comfortable owning a project from concept through deployment.

TECHNICAL SKILLS

Programming	Python, C/C++, MATLAB, Ladder Logic, Structured Text, URScript, Epson SPEL+	Systems	Studio 5000, PLC programming, robotic arm integration, embedded firmware
Protocols	EtherNet/IP · CIP, MQTT, UART / SPI / I2C, Modbus TCP/IP, USB	Design	Altium Designer, KiCAD, Fusion 360, SOLIDWORKS, AutoCAD, Simulink, Automation Studio
Libraries	pylogix, pygame, OpenCV, NumPy / SciPy, ESP-IDF	Lab	Custom PCB design, soldering & rework, oscilloscopes, multimeters, signal generators, manual machining, 3D printing

SELECTED PROJECTS

Modular Industrial Educational Platform - Senior Capstone Component

2026

Rockwell Studio 5000 · Allen-Bradley CompactLogix · EtherNet/IP · UR5e · Python · pylogix · HMI

Sole engineer; designed, built, and integrated a portable industrial automation demo in under three weeks - EtherNet/IP network linking an Allen-Bradley CompactLogix PLC, remote I/O, and UR5e, manual robotic control with PS5 controller via Python

Industrial Educational Platform - Senior Capstone Project

2025

Rockwell Studio 5000 · EtherNet/IP · UR3 · Epson SCARA · Pneumatics · Fusion 360

Controls engineer on a Manufacturing Education Alliance sponsored capstone. Programmed the PLC, configured EtherNet/IP networking and remote I/O, and integrated two UR3s, an Epson SCARA, conveyors in Studio 5000.

Hemiparesis Neglect Reminder - Dignity Health Sponsored

2026

Raspberry Pi 5 · MediaPipe · Python · Fusion 360 · 3D Printing

Mechanical lead on a Dignity Health-sponsored therapy device for hemiparesis patients. Designed and 3D-printed the enclosure in Fusion 360; system runs MediaPipe facial tracking on a Raspberry Pi 5 to detect head pose, deliver audio prompts, and log session data for clinician review.

PROFESSIONAL EXPERIENCE

Principle Service Solutions

Oct 2020 - Jun 2022

Engineering Maintenance Technician - contracted to Hitachi High Tech / Intel, Chandler AZ

- Performed routine and emergency maintenance on Hitachi dry-etch process tools in a Class 1 cleanroom at Intel; diagnosed and resolved electrical and mechanical faults across vacuum, RF, robotic, and gas-delivery subsystems.

Phoenix Test Arrays

May 2020 - Oct 2020

Manufacturing Technician - Phoenix, AZ

- Manufactured precision test contacts to tight dimensional tolerances; performed electrical verification on completed assemblies.

Earlier roles (2015–2020): VoIP Support Technician, J Curve Networks · Paramedic / EMT, Lifeline / PMT Ambulance · Medical Staff Associate, CSL Plasma

EDUCATION

Arizona State University Polytechnic - Mesa, AZ

Completed Spring 2026

B.S.E., Engineering (Robotics)

Mesa Community College - Mesa, AZ

Completed Spring 2020

Associate of Science